2025 28th International Conference on Computer and Information Technology (ICCIT)
19-21 December 2025, Cox's Bazar, Bangladesh

Pothole Detection in Various Rainy Conditions Using YOLOv11: Leveraging Synthetic Augmentation and Attention Techniques

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***Abstract*—Potholes are a major problem on roadways, causing dangerous accidents and raising the maintenance costs of roads for transportation authorities. Adverse conditions such as rain, fog, and nighttime degrade sensor performance and obscure road images, posing significant challenges for accurate pothole detection. Among these, rainy weather is one of the most challenging conditions, introducing many complex sensing difficulties, including low visibility, water in potholes, reflectiveness of wet surfaces and raindrops on camera lenses, and degrading detection performance of detection systems. To address this, we propose an approach that focuses on enhancing pothole detection specifically under rainy conditions by fine-tuning the YOLOv11 with the Convolutional Block Attention Module (CBAM). This integration improves spatial and channel-wise feature representation, enabling better performance in visually challenging conditions. To enhance model interpretability, Grad-CAM++ is utilized to highlight the critical regions influencing the model's detection decisions. The model is trained on a Synthetic Rain Weather Augmented dataset, which includes a wide range of rain effects and intensities, thereby improving its robustness in real-world rainy environments. The resulting model achieved a precision of 0.881, a recall of 0.755, a mAP_{50} of 0.819, and a mAP_{50-95} of 0.575, demonstrating strong performance under challenging weather conditions.**

***Keywords*—Pothole detection, road safety, YOLOv11, Convolutional Block Attention Module (CBAM), Grad-CAM++, Synthetic Rain Augmentation**

I. Introduction

Autonomous vehicles aim to provide safer, more efficient transportation by independently navigating complex environments. However, AVs still face significant challenges in adverse weather conditions such as rain, fog, and nighttime. Rain makes pothole detection challenging for these vehicles by filling potholes with water, creating reflections on wet surfaces, and leaving raindrops on camera lenses, all of which can mask a pothole's boundaries. Although LiDAR is effective, enhancing detection through cameras offers a more cost-efficient solution.

The condition of roadways greatly affects vehicle safety and performance, particularly impacting tires and suspension systems. When roads contain extensive potholes and cracks, vehicles suffer increased wear and are prone to accidents. The severity of these road defects determines the urgency and level of maintenance required. Assessing and categorizing the severity of potholes and similar damages is essential for effective maintenance planning. Such severity assessments help authorities make informed decisions regarding repair priorities, resource allocation, and possible road closures to prevent additional vehicle damage and ensure safer travel conditions.

In this study, we employ advances in AI and computer vision by integrating the Convolutional Block Attention Module (CBAM) into YOLOv11, a real-time detection framework. This combination enhances feature representation, making it more effective in rainy environments. Grad-CAM++ further aids by providing visual explanations of model predictions. The system was trained on a Synthetic Rain Weather Augmented dataset simulating various levels of rain severity, which enhances the model's robustness across a wide range of real-world conditions. The final dataset consisted of 10,940 images with 32,560 labeled potholes, including both original and rain-augmented images.

This paper is structured as follows. Related work is reviewed in Section II. The proposed methodology is presented in Section III. The experimental work and corresponding results are detailed in Section IV. Finally, the conclusion and suggestions for future research are provided in Section V.

II. Related Work

Pothole detection is an active research area in intelligent transportation systems, where deep learning plays a central role. The YOLO family has seen significant advancements in object detection, being known for their efficiency as well as the ability to work in real-time. YOLOv5, known for its speed and ease, achieved around 80% mAP on clear road images, like those in Kaggle's pothole dataset, but struggled in wet weather, and its performance dropped to 65% [1]. YOLOv7 and YOLOv8 pushed the bar higher, with YOLOv8 reaching 85% mAP in good conditions thanks to its upgraded backbone and anchor-free detection [12]. However, these models work great when the sun's out, but rain throws them off, dropping accuracy due to water-filled potholes and reflections. We chose YOLOv11 because it's a step ahead, designed to handle tough conditions like rain with a larger, more powerful architecture. Its improved backbone and attention mechanisms make it better at spotting small or obscured potholes, which is critical for wet roads. Recent benchmarks show that YOLOv11 variants

979-8-3315-7867-1/25/31.00©2025IEEE



(a) Clear weather (b) Light Afternoon Rain (c) Evening Drizzle (d) Heavy Rain in Evening (e) Heavy Rain with Storm

Fig. 1: Views of augmented data under various rainy conditions.

outperform previous models, especially in speed and reliability, even in real-time scenarios. YOLOv11 integrates well with our rain-augmented dataset generated using OpenCV. We also added a Convolutional Block Attention Module (CBAM) to help the model focus on important pothole features, like edges, even in low visibility during rain. Combined with Grad-CAM++, improves interpretability and robustness in rainy conditions. This addresses a significant gap in research and application.

III. Methodology

This section describes our approach, focusing on the novel integrations and tailored enhancements of existing models. An advanced real-time object detection framework, YOLOv11, serves as the core detection model, enhanced with a Convolutional Block Attention Module (CBAM) and fine-tuned for robustness under various rainy conditions. Additionally, Gradient-weighted Class Activation Mapping (Grad-CAM++) is integrated to provide interpretability. Below, we outline our contributions, particularly emphasizing dataset creation, model adaptation with CBAM, fine-tuning strategies, and interpretability analysis.

A. Datasets

In this work, we utilize a combination of real-world and synthetic datasets to enhance pothole detection performance under varying weather conditions. The dataset consists of 2,188 annotated images collected from RoboFlow [7] primarily sourced from real-world South Asian roads, including regions in India, Bangladesh, and Pakistan. These images were captured using smartphones, high-resolution cameras, and vehicle-mounted dashcams, representing a diverse range of road environments.

While the original dataset primarily contains clear weather images, real-world pothole detection systems must operate under challenging environmental conditions such as rain, fog, and low visibility. To address this limitation, we introduce a custom dataset of 8,752 weather-augmented images with simulated rain effects, generated using OpenCV. The combined dataset totals 10,940 images with 32,560 labeled objects. These augmentations cover a variety of realistic rainy scenarios, including light afternoon rain, evening drizzle, heavy rainfall in the evening, and heavy rain with storm conditions, and incorporate visual effects such as obscured views, glare, or rain interference scenarios often absent in public datasets, yet critical for real-world applications such as autonomous driving and infrastructure monitoring.

OpenCV, the Open Source Computer Vision Library, is a powerful and widely used open-source library designed for real-time computer vision and image processing tasks. In this study, OpenCV was used to simulate realistic rainy weather conditions for data augmentation in a pothole detection dataset. Techniques included adding atmospheric haze to mimic depth, drawing angled and blurred lines for rain streaks, and blending flipped reflections to create water puddles. Fog effects were generated using soft overlays with variable density, motion blur simulated camera shake, and raindrops were created using blurred circular masks. Lighting adjustments further enhanced realism, improving model robustness in adverse conditions.

Potholes in Light Afternoon Rain: The Light Afternoon Rain simulation emulates light rainfall under daylight conditions. Using OpenCV [2], a mild brightness reduction (8%) and slight enhancement of the red channel were applied to replicate warm afternoon lighting. Atmospheric haze with an intensity of 0.12 simulated depth, while water puddles were introduced through blended flipped reflections at 0.15 intensity. Light rain streaks were overlaid using randomly angled, blurred lines, followed by a light fog layer and a few soft lens raindrops. A total of 2188 images were generated, distributed as 1531 training, 438 validation, and 219 test samples.

Potholes in Evening Drizzle: The Evening Drizzle Conditions simulation was designed to reflect early evening or cloudy scenarios with reduced ambient light. Using OpenCV, the image brightness was reduced to 70%, and atmospheric haze was applied with an intensity of 0.20. Reflection intensity for puddles was increased to 0.25, while fog density ranged moderately with a Gaussian spread. Light rain streaks, motion blur ,and 8 circular raindrop overlays were added to enhance realism. These settings preserved important visual structures while introducing noticeable weather degradation [10]. A total of 2188 images were generated and split into 1531 training, 438 validation, and 219 test samples.

Potholes in Heavy Rainfall in Evening: Heavy rainfall under low-light evening conditions was simulated by darkening images to 60% brightness, reducing red-green channel

TABLE I: Distribution of custom and Roboflow datasets

Dataset		Train	Valid	Test	Total
Original Dataset		1531	438	219	2188
Our Custom Dataset	Light Afternoon Rain	1531	438	219	2188
	Potholes in Evening Drizzle	1531	438	219	2188
	Heavy Rainfall in Evening	1531	438	219	2188
	Heavy Rain with Storm	1531	438	219	2188
Total		7655	2190	1095	10940

intensity, and adding cool tones [8]. Atmospheric haze , strong puddle reflections, and moderately dense rain streaks were added. Fog and motion blur enhanced depth and scene degradation, and 12 raindrops simulated lens interference. These transformations, achieved using OpenCV, preserve semantic structure while introducing substantial variation. The output includes 2188 images, split across 1531 training, 438 validation, and 219 test samples.

Potholes in Heavy Rain with Storm: Simulating extreme storm conditions, this augmentation darkened images to 50%, with channel adjustments to produce a stormy tone. Atmospheric haze of 0.28, puddle reflections at 0.4, and dense, steeply angled rain streaks were applied. Strong motion blur and heavy fog layers obscured much of the background, simulating poor visibility. 20 blurred circular raindrops were added to emulate water droplets on the lens [11]. Generated using OpenCV, this scenario reflects the most degraded visual condition, producing 2188 samples across 1531 training, 438 validation, and 219 test sets.

Table I shows the distribution of our custom dataset compared with the original Roboflow road pothole dataset.

B. You Only Look Once version 11 (YOLOv11) Model

YOLOv11 is the latest release in the YOLO family of single-stage object detectors, offering significant improvements in both accuracy and efficiency through architectural innovations [6]. This makes it particularly effective for tasks like pothole detection in adverse weather conditions, where visibility is low and features are partially obscured.

- **Backbone Enhancements:** YOLOv11 introduces the C3k2 module (a Cross Stage Partial block with double kernel convolution), the SPPF (Spatial Pyramid Pooling – Fast), and C2PSA (Convolution with Parallel Spatial Attention). These components allow the model to extract both fine-grained and global contextual features efficiently, which is essential when detecting degraded or partially visible pothole structures under rain or haze.
- **Neck with Attention Mechanism:** The neck architecture builds on multi-scale feature fusion using concatenation, upsampling, and PSA-enhanced blocks. This enables better detection of potholes of varying sizes, especially in low-contrast regions caused by atmospheric distortions like fog or rain streaks.
- **Detection Head:** While retaining the proven YOLO-style detection head, YOLOv11 benefits from the improved upstream feature maps. This contributes to more accurate bounding box localization and class confidence scores, even when weather effects obscure clear object boundaries.
- **Anchor-Free Design:** YOLOv11 adopts an anchor-free mechanism, directly predicting object centers and dimensions. This approach adapts well to the irregular and non-uniform shapes of real-world road damages, such as cracks and potholes.
- **Multi-Task Capability & Model Scalability:** Supporting tasks beyond detection (e.g., instance segmentation, pose estimation, and oriented bounding box detection), YOLOv11 scales across nano to extra-large variants, offering flexibility across deployment settings—from cloud-based to edge devices.
- **Efficiency-Accuracy Tradeoff:** YOLOv11's architecture achieves an optimal balance through lightweight yet expressive modules. In real-time detection tasks, especially under visually degraded conditions, it maintains strong performance with reasonable computational demands.

Overall, YOLOv11 fuses spatial attention, efficient feature fusion, and a simplified detection strategy to handle complex image scenarios. Its ability to capture subtle variations and maintain performance under visual noise makes it a highly suitable detector for our rain-augmented, low-visibility pothole dataset.

C. Fine-Tuning with Convolutional Block Attention Module (CBAM)

To further enhance YOLOv11's feature learning under rainy conditions, we integrated a Convolutional Block Attention Module (CBAM) into the pretrained YOLOv11 model. CBAM sequentially applies channel attention and spatial attention to intermediate feature maps, helping the network focus on what and where to attend within visually difficult scenes. In our approach, CBAM was specifically integrated into Layer 5 and Layer 11 of the model. Layer 5, responsible for mid-level feature extraction, captures complex textures, while Layer 11 processes upsampled features, refining spatial context for better detection. This refinement improves detection in areas affected by rain streaks, puddles, or glare without altering the base detection head. The YOLOv11 model with CBAM was fine-tuned on the rain-augmented dataset using a smaller learning rate. This enabled the model to retain its pretrained knowledge while adapting to new noise patterns induced by rainy conditions. This lightweight integration improves both robustness and interpretability with minimal computational overhead [9].

D. Interpretability with Grad-CAM++

To assess and interpret the decision-making of the fine-tuned YOLOv11+CBAM model, we employed Grad-CAM++ [3], an advanced visualization technique. Grad-CAM++ highlights spatial regions in the image that contribute most to the model's detection confidence. These heatmaps were overlaid on rain-affected inputs, allowing qualitative verification that the model correctly focuses on pothole regions rather than rain artifacts or irrelevant textures. This not only supports

model transparency but also validates the attention alignment introduced by CBAM.

E. Evaluation

We will evaluate the performance of our proposed approach based on its precision, recall and F1-score.

Precision (P): It is calculated by dividing the number of True Positives (TP) by the total number of True Positives and False Positives (FP). It is expressed mathematically as follows:

$$\text{Precision}(P) = \frac{TP}{TP + FP}.$$

Precision is a performance metric used in classification tasks to measure the accuracy of positive predictions

Recall (R): Recall is calculated as the ratio of True Positives (TP) to the sum of True Positives and False Negatives (FN), and is mathematically defined as follows:

$$\text{Recall}(R) = \frac{TP}{TP + FN}.$$

Recall reflects the model's ability to capture all relevant positive cases, whereas precision indicates the correctness of its positive predictions.Recall values that are low indicate a high number of false negatives.

F1- Score(F1) : The product of precision and recall is divided by the total of the precision and recall results in half of the F1 score.It is the harmonic mean of precision and recall. Mathematically, it is defined as:

$$\text{F1-Score}(F1) = 2 \times \frac{P * R}{P + R}.$$

IV. Results and Discussion

We trained our YOLOv11 model on a dataset with different weather conditions for 150 epochs. The training and validation results from the first training process are shown in Fig.2, and then we fine-tune the model by incorporating CBAM and training for another 50 epochs, whose results are demonstrated in Fig.3. After fine-tuning, the model evaluation comes up with a precision of 0.881, a recall of 0.755, a map_{50} of 0.819, and a map_{50-95} of 0.575.

The performance of the YOLOv11+CBAM model was assessed through several key metrics, starting with its confidence-based behavior. As shown in the Precision-Confidence Plot in Fig 5a, the model reaches a peak precision of 1.00 when the confidence threshold is set to 0.95, indicating that at high confidence, all detected potholes are accurate with no false positives. The Recall-Confidence Plot in Fig. 5b reveals a maximum recall of 0.94 at a confidence of 0, suggesting the model captures nearly all potholes at lower thresholds, though with some false positives included. The F1-Confidence Plot in Fig. 5c highlights an optimal F1 score of 0.81 at a 0.56 confidence level, striking a balanced performance between precision and recall.

Overall accuracy is further evaluated via the Precision-Recall Graph in Fig. 5d, where the YOLOv11+CBAM model achieves an mAP_{50} of 0.819, reflecting strong detection capability at an IoU of 0.5. The Confusion Matrix in Fig. 6 provides additional insight, with a true positive rate of 0.79 for potholes and a true negative rate of 1.00 for non-pothole areas, showing reliable classification performance.

Table II compares the YOLOv11 and YOLOv11+CBAM models against other approaches. Faster R-CNN, relying on real-world data, lags with a precision of 53.4%, recall of 45.2%, and mAP_{50} of 45.5%. YOLOv8l, enhanced with augmented data, improves to 63.8% precision and 52.7% recall but falls short in mAP_{50-95} at 17.2%. YOLOv9, also using augmented data, performs well with 80.7% precision, 75.9% recall, and an mAP_{50} of 82.9%. The base YOLOv11 model achieves 84.2% precision, 77.0% recall, and 56.0% mAP_{50-95}, while fine-tuning with CBAM pushes the performance to 88.1% precision, 75.5% recall, and 81.9% mAP_{50}, with an mAP_{50-95} of 57.5%, outperforming others in consistency across IoU levels.

The models were trained on a dataset blending synthetic images mimicking weather challenges with real-world pothole data. The CBAM fine-tuning, applied after initial training, enhances the model's adaptability, making it a promising tool for real-time road monitoring and autonomous driving systems in varied conditions.And then, to assess and interpret the decision-making of the fine-tuned YOLOv11+CBAM model, we employed Grad-CAM++,an advanced visualization technique.

V. Conclusion

Damages in the road such as potholes and cracks are examples of costs related to road safety. These costs are necessary to take into account since they can negatively affect the health of the vehicles and their tires. In this paper, we utilized Yolov11 as a base detection model enhanced with a Convolutional Block Attention Module (CBAM) and fine-tuned for robustness under various rainy conditions. Additionally, Grad-CAM++ is integrated to provide interpretability. Due to the absence of publicly available real rainy-condition pothole datasets, our evaluation under rain relies mainly on synthetic augmentation. As a result, the model has not been tested on completely unseen potholes in natural rain. Despite these limitations, the model shows promising performance under challenging rainy conditions. In the future, we plan to collect real rainy-condition data, conduct full ablation studies, and explore domain adaptation and additional adverse weather scenarios such as fog, snow, and nighttime to improve real-world robustness.

VI. Acknowledgement

The authors would like to acknowledge the support provided by the supervisor, parents, and International Islamic University Chittagong (IIUC).

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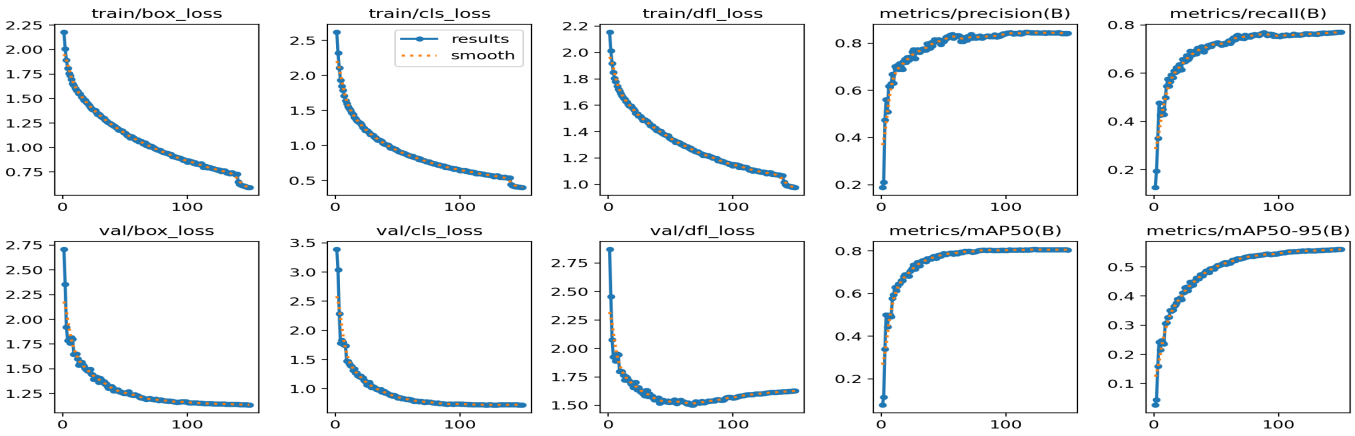


Fig. 2: The training/validation results of the trained YOLOv11 model on our generated dataset for 150 epochs.

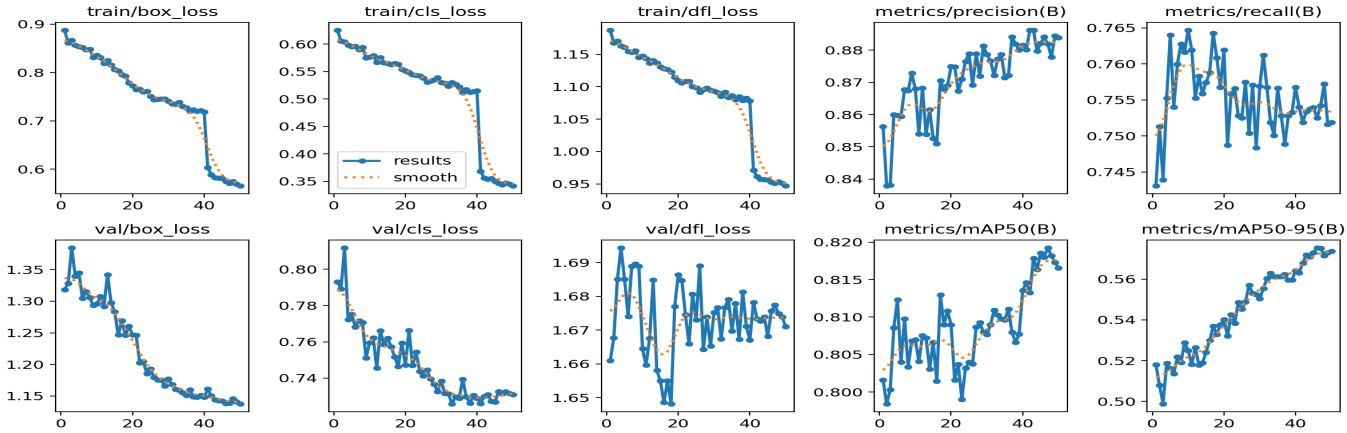
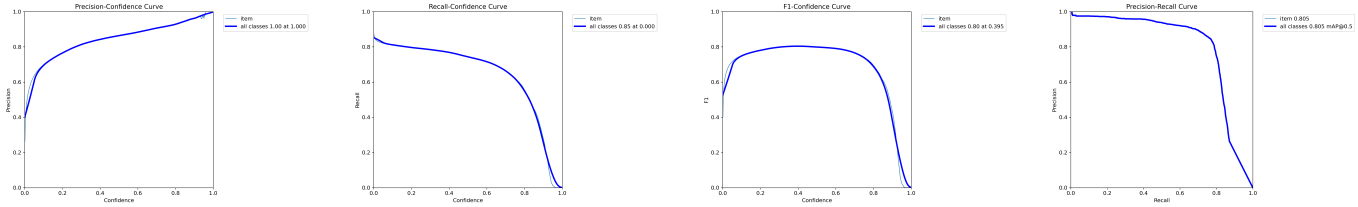
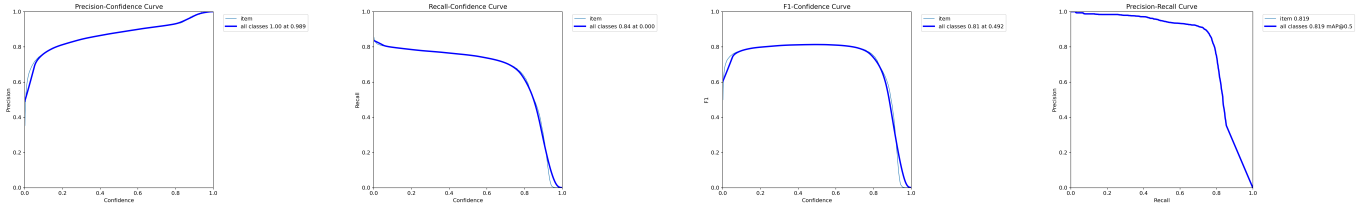


Fig. 3: The training/validation results of the trained YOLOv11 model and enhanced with CBAM on our generated dataset for 50 epochs.



(a) Precision curve. (b) Recall curve. (c) F1 curve. (d) Precision-recall curve.

Fig. 4: Results of the trained YOLOv11 model on our generated dataset for 150 epochs.



(a) Precision curve. (b) Recall curve. (c) F1 curve. (d) Precision-recall curve.

Fig. 5: Results of the trained YOLOv11 enhanced with the CBAM model on our generated dataset for 50 epochs.

TABLE II: Comparative results. Baseline metrics from papers with limited rainy data and different datasets; values are not directly comparable.

Model	Dataset Type	Precision	Recall	F1-Score	mAP@0.5	mAP@0.5:0.95	Ref.	Year
Faster R-CNN	Real-world dataset	53.4%	45.2%	48.9%	45.5%	15.9%	[5]	2023
YOLOv8l	Real + Augmented	63.8%	52.7%	–	42.6%	17.2%	[12]	2024
YOLOv9	Real + Augmented	80.7%	75.9%	78.0%	82.9%	55.1%	[4]	2025
YOLOv11	Real + Augmented	84.2%	77.0%	–	80.5%	56.0%	Our work	–
YOLOv11+CBAM	Real + Augmented	88.1%	75.5%	81.3%	81.9%	57.5%	Our work	–

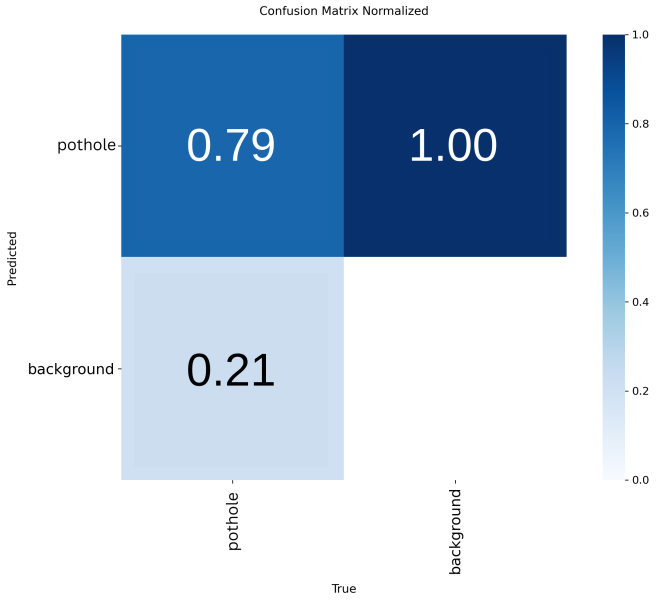
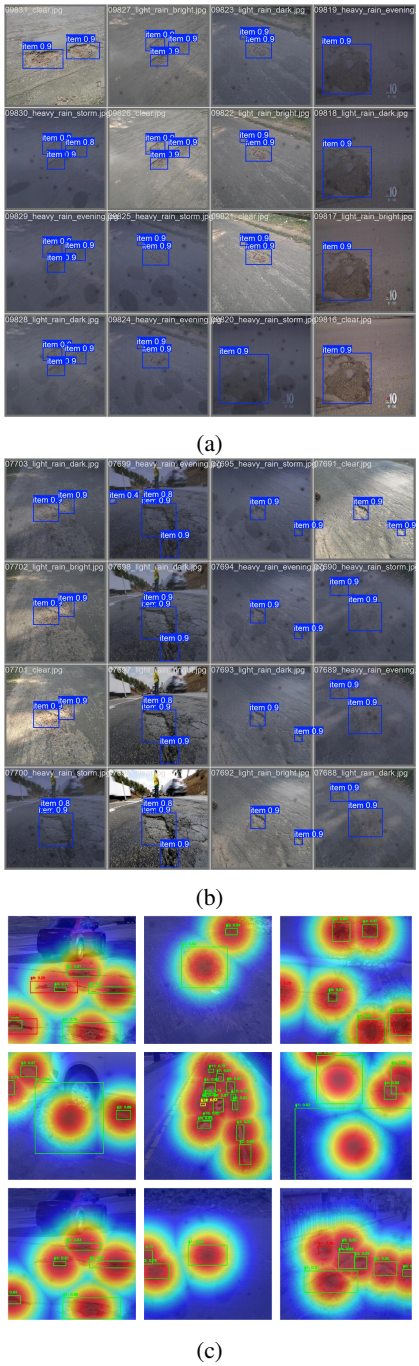


Fig. 6: Confusion matrix of the trained YOLOv11+CBAM model for 50 epochs.



(a) (b) (c)

Fig. 7: Some detection attempts on our generated dataset.

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